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FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VII

JEE (Main)-2024

TEST DATE: 22-03-2024

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

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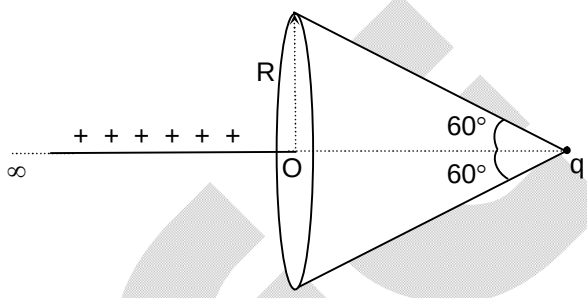
Physics

PART – A

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

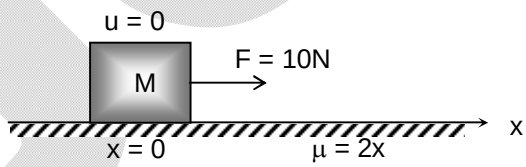
1. Figure shows a semi-infinite uniformly charged wire along axis of an imaginary cone such that its end coincides with centre O of base of cone. A point charge q is placed at the apex of the cone. If electric flux through the base of cone is zero, then the linear charge density of wire is

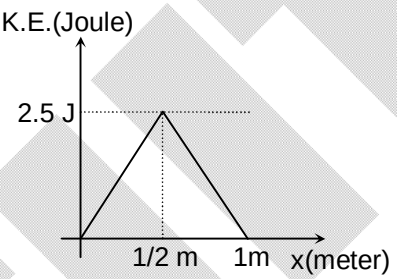


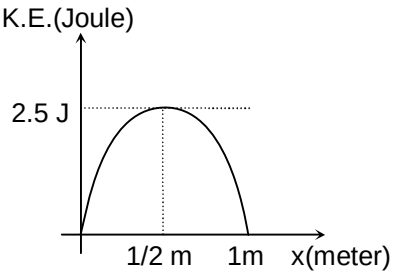
(A) $\frac{q}{4R}$ (B) $\frac{q}{2R}$

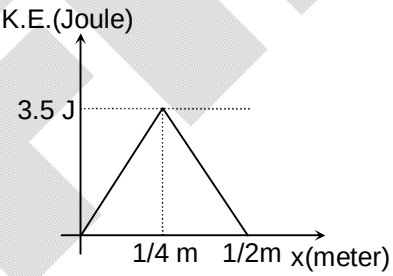
(C) $\frac{q}{8R}$ (D) $\frac{q}{R}$

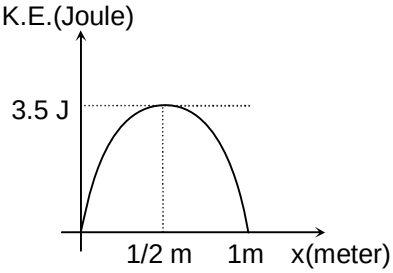
2. A small block of mass M = 1kg is placed on a rough horizontal surface having coefficient of friction varies with distance as $\mu = 2x$. The block is pulled by constant horizontal force F = 10 N from x = 0 as shown in the figure. Which of the graph showing the variation of kinetic energy of block with position x.



(A)


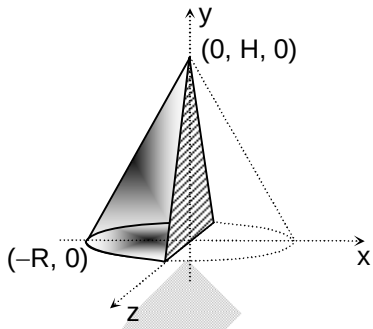
(B)


(C)


(D)


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3. The coordinate of centre of mass of half uniform solid cone shown in the figure is



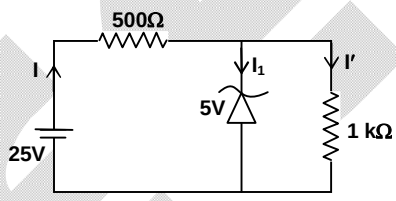
(A) $\left(-\frac{2R}{\pi}, \frac{H}{4}\right)$

(B) $\left(-\frac{4R}{\pi}, \frac{H}{3}\right)$

(C) $\left(-\frac{R}{\pi}, \frac{H}{4}\right)$

(D) $\left(\frac{2R}{\pi}, \frac{H}{4}\right)$

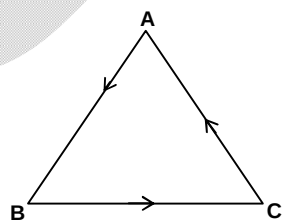
4. The ratio of current I supplied by the cell to current I_1 in the Zener diode having break down voltage V_z = 5V as shown in the figure, is



(A) $\frac{7}{8}$ (B) $\frac{6}{7}$

(C) $\frac{8}{7}$ (D) $\frac{5}{7}$

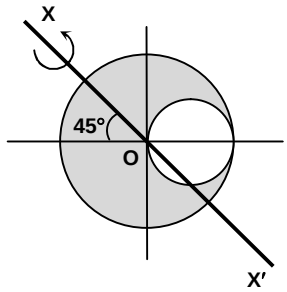
5. Three point particles A, B and C initially located at vertices of an equilateral triangle of side ℓ start moving with constant speed v each such that A always moves towards B, B always moves towards C and C always moves towards A. The average velocity of A for its motion till the instant it catches B is



(A) $\frac{v}{2\sqrt{3}}$ (B) $\frac{v}{\sqrt{3}}$

(C) $\frac{\sqrt{3}v}{2}$ (D) $\frac{v}{2}$

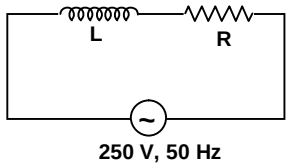
6. A disc of radius $\frac{R}{2}$ is cut from a uniform disc of radius R and surface mass density σ as shown in the figure, the moment of inertia of the remaining part about an axis XX′ which passes through the centre O and is in the plane of the disc will be



(A) $\frac{19}{64}\sigma\pi R^4$ (B) $\frac{13}{22}\sigma\pi R^4$

(C) $\frac{13}{64}\sigma\pi R^4$ (D) $\frac{13}{16}\sigma\pi R^4$

7. A series L-R circuit is connected across a source of 250 V, 50 Hz supply. The power factor of the circuit is 0.8 and power of 400 W is dissipated. The value of inductive reactance and resistance are



(A) $X_L = 75\ \Omega,\ R = 100\ \Omega$

(B) $X_L = 1000\ \Omega,\ R = 75\ \Omega$

(C) $X_L = 70\ \Omega,\ R = 1000\ \Omega$

(D) $X_L = 50\ \Omega,\ R = 1000\ \Omega$

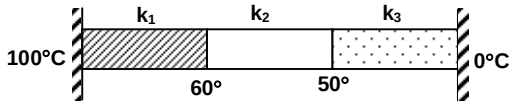
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8. A heavy nucleus at rest breaks into two nuclei which move away with velocities in the ratio of 1 : 8, the ratio of the radii of the nuclei (Assumed to be spherical) is

(A) 1 : 3 (B) 2 : 1

(C) 3 : 2 (D) 2 : 3

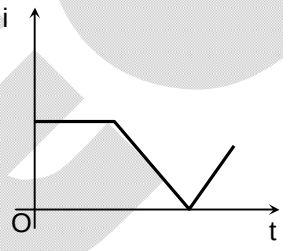
9. Figure shows three rods of identical cross-section and length made up of different materials of thermal conductivity k_1, k_2 and k_3 respectively joined together to make a long rod. The ends of the rod are maintained at temperature 100°C and 0°C. If the joints are at temperature 60°C and 50°C under steady state, then k_1, k_2 and k_3 are related as

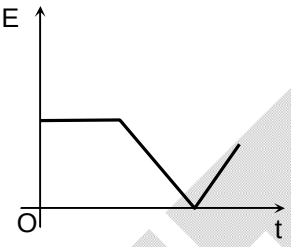


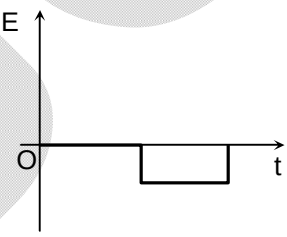
(A) $k_1 = k_2 = 2k_3$ (B) $4k_1 = k_2 = 5k_3$

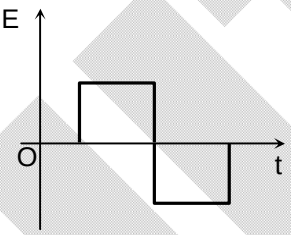
(C) $2k_1 = 3k_2 = 4k_3$ (D) $k_1 = k_2 = k_3$

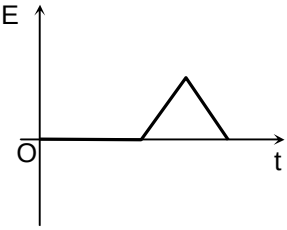
10. The current i in an inductor coil varies with time t according to the graph as shown in the figure. Which of the following graph shows the induce emf(E) in the coil with time



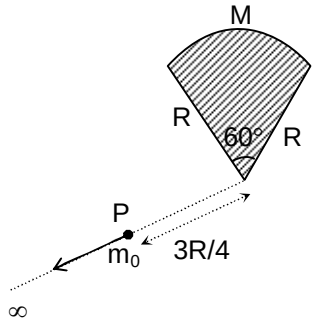
(A)


(B)


(C)


(D)


11. Figure shows a uniform sector lamina of radius R and mass m making an angle 60° at its centre. A point mass m_0 is placed at a distance $\frac{3R}{4}$ from centre on the axis. The work done in slowly moving m_0 from initial point P to infinity is

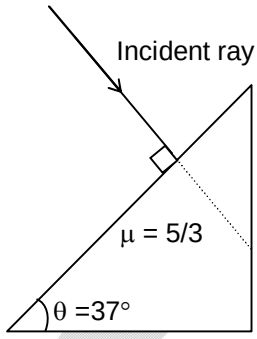


(A) $-\frac{6GMm_0}{R}$ (B) $-\frac{4GMm_0}{R}$

(C) $\frac{GMm_0}{R}$ (D) $\frac{2GMm_0}{R}$

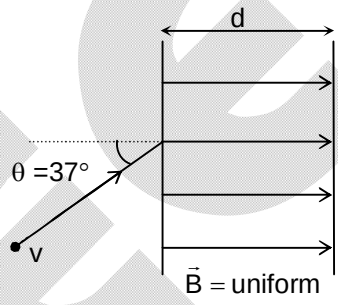
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12. A light ray incident perpendicularly on a face of a right angle prism as shown in the figure. The angle of deviation is.... $\left(\sin 37° = \frac{3}{5}\right)$



(A) 74°
(B) 106°
(C) 127°
(D) 53°

13. A proton moving with speed v = 4×10^5 m/s enters an region of magnetic field B = π T with velocity making angle θ = 37° with field. The proton completes 15 revolution by the time it crosses the region of width 'd'. The value of d is (Take mass of proton = 1.6×10^{-27} kg, charge of proton = 1.6×10^{-19} C)



(A) 55 mm
(B) 96 mm
(C) 40 mm
(D) 30 mm

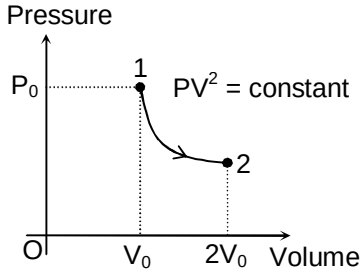
14. Light of energy 2eV and intensity 8 W/m^2 falls on a metal surface of area 10 cm^2 and work function 1.5 eV. Only 20% of incident photons produce photoelectrons. The number of emitted photoelectron per second will be

(A) 4×10^{17}
(B) 5×10^{17}
(C) 5×10^{15}
(D) 5×10^{18}

15. If work done by gas molecule is given by the expression $W = \alpha\beta e^{-\frac{\beta x^2}{k\theta}}$; where x is the displacement, k is the Boltzmann constant and θ is temperature. If α and β and constant. Dimensional formula of α will be

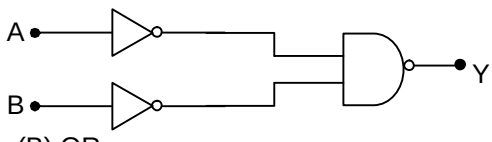
(A) $\left[M^0L^2T^0\right]$
(B) $\left[M^0LT^0\right]$
(C) $\left[MLT^{-1}\right]$
(D) $\left[M^2LT^{-2}\right]$

16. An ideal monoatomic gas undergoes a thermodynamics process form 1→2 as shown in the figure. The change in internal energy is



(A) $\frac{3}{4}P_0V_0$
(B) $-\frac{3}{4}P_0V_0$
(C) $\frac{3}{2}P_0V_0$
(D) $-\frac{3}{2}P_0V_0$

17. Identify the logic operation performed by the given circuit



(A) NAND
(B) OR
(C) NOR
(D) AND

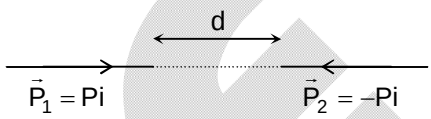
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18. Time period of a simple pendulum with point of suspension stationary is T_1. Now the suspension point moved upward according to the equation $y = 2t^2$ where y is the vertical displacement and time period becomes T_2. The ratio of $\frac{T_2^2}{T_1^2}$ is (take g = 10 m/s^2)

(A) $\frac{5}{6}$ (B) $\frac{7}{5}$

(C) $\frac{5}{7}$ (D) $\frac{6}{5}$

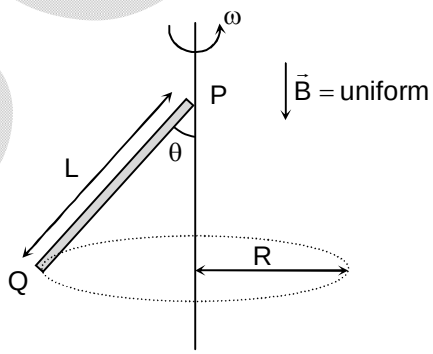
19. Two short electric dipole each of mass m having dipole moment $\vec{P}_1 = P\hat{i}$ and $\vec{P}_2 = -P\hat{i}$ are held on the x-axis at a distance d from each other. When released they move away from each other without changing the direction of dipole moment. The speed of each dipole when separation between dipoles becomes 2d is



(A) $\frac{P}{2d}\sqrt{\frac{7}{\pi\varepsilon_0 md}}$ (B) $\frac{P}{4d}\sqrt{\frac{7}{\pi\varepsilon_0 md}}$

(C) $\frac{P}{8d}\sqrt{\frac{7}{\pi\varepsilon_0 md}}$ (D) $\frac{P}{d}\sqrt{\frac{7}{\pi\varepsilon_0 md}}$

20. Figure shows a rod PQ whose one end is rotated on a circular path of radius R. Angular speed of rotation is ω. There exists a uniform magnetic field B downward. Then induce emf in rod is



(A) $\frac{1}{2}B\omega R^2 \sin^2\theta$

(B) $\frac{1}{2}B\omega L^2 \sin^2\theta$

(C) $\frac{1}{2}B\omega L^2 \cos^2\theta$

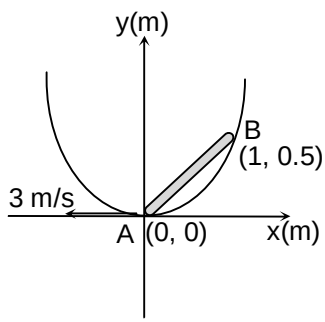
(D) $\frac{1}{2}B\omega R^2 \cos^2\theta$

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

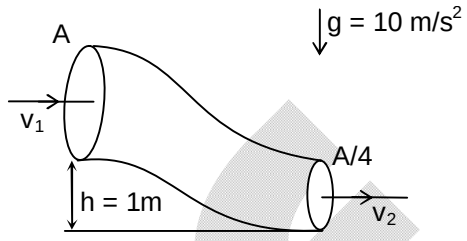
21. A rod moving with its end A and B in contact with a parabolic surface $y = \frac{x^2}{2}$ as shown in figure. At the instant shown, coordinates of A and B are given and velocity of end A is 3 m/s. Calculate the angular velocity (in rad/s) of rod at the given moment.



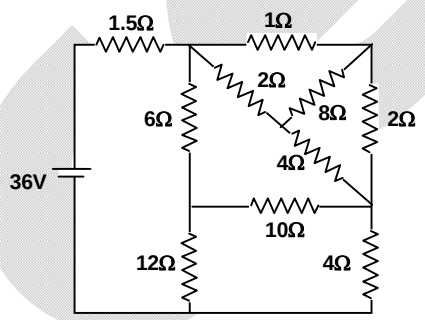
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22. In Young's double slit experiment, a monochromatic light of wavelength λ is used. The intensity of light at a point on screen, where the path difference is λ is I_0 units. The intensity of light on the screen where the path difference $\frac{\lambda}{4}$ is given by $\frac{I_0}{n}$ units. Find the value of n.

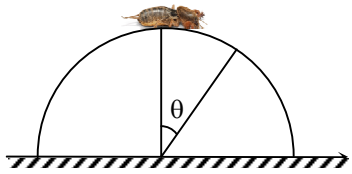
23. A non-viscous liquid of density 800 kg/m^3 flows in streamline motion through a tube that tapers in cross-section area from A to $\frac{A}{4}$. The pressure difference between the wide and narrow section of pipe is 10375 Pa. At narrow section, the velocity (in m/s) of fluid is ……..



24. Calculate current in ampere through the resistor R = 1.5Ω?



25. An insect crawls from top of a fixed hemispherical surface of radius $\frac{24}{\pi}$ cm very slowly. The coefficient of static friction between the insect and surface is $\frac{1}{\sqrt{3}}$. Find the distance (in cm) travelled by the insect before slipping.

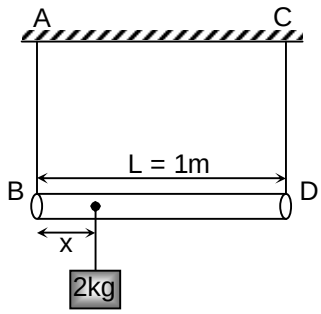


26. The electric field of a plane electromagnetic wave is given by (speed of light c = 3×10^8 m/s)

$$E = 100\sin\left[2\pi \times 2 \times 10^{15}\left(t - \frac{x}{c}\right)\right] \text{ V/m}$$

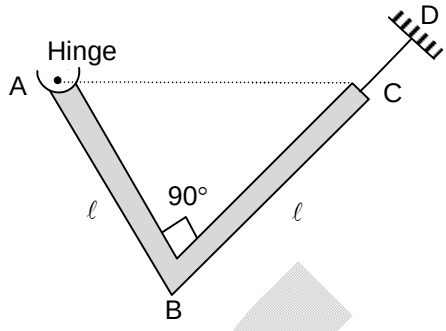
If this wave falls normally on a perfectly reflecting surface having an area of 10 cm^2. The radiation pressure exerted by the electromagnetic wave on the surface during a 10 minute of exposure is $\frac{x}{10^{10}}$ N/m^2. Fin the value of x. (premittivity of free space $\varepsilon_0 = 8.85 \times 10^{-12}$ C^{2}N^{-1}m^{-2})

27. A massless rod of length L = 1m is suspended by two identical string AB and CD of equal length. A block of mass 2 kg is attached at a distance x from end B as shown in the figure. The frequency of 1st harmonic in AB is equal to 2nd harmonic in CD. The value of x (in cm) is (g = 10 m/s^2)

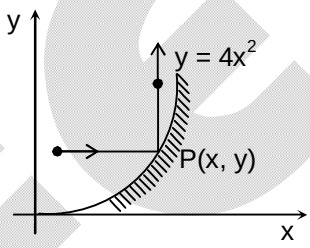


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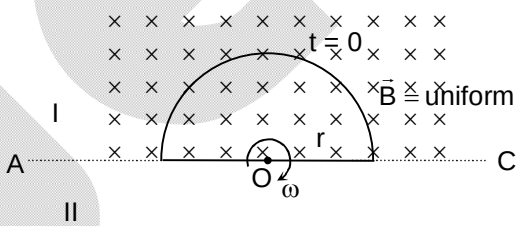
28. A uniform L-shaped rod of length $2\ell = \frac{6}{\sqrt{2}}$ m (AB= BC = ℓ) is hinged at one end A and other end C is connected with string to wall D as shown in the figure. The string CD is cut. Calculate angular acceleration (radian/s^2) of rod just after the string CD is cut ?



29. A ball moving horizontally strikes a smooth parabolic surface given by $y = 4x^2$ at point P(x,y). Coefficient of restitution for collision is $e = \frac{1}{4}$. If the ball moves vertically after collision, the value of $y = \frac{1}{n}$. Calculate the value of n.



30. Figure shows a semicircular conducting loop of radius r and resistance R is made to rotate with constant angular speed ω about its centre O. Space is divided into two regions by the line AC. There exists uniform magnetic field perpendicular to the plane in region I. The magnitude of charge flow in the interval t = 0 to $t = \frac{\pi}{2\omega}$ is $\frac{B\pi r^2}{nR}$. Find the value of n.



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Chemistry

PART – B

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. A solution of NH_4Cl and NH_4OH has pH = 9, which of following compounds could be precipitated when this solution is added in equal volume to a solution containing 0.2 M of metal ions.

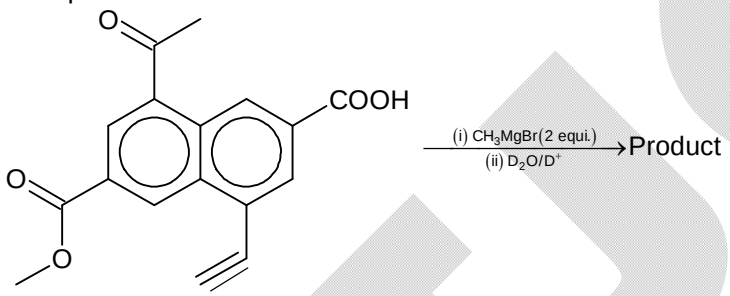
(A) AgOH $\left(K_{sp} = 3\times10^{-4}\right)$

(B) $Ca(OH)_2$ $\left(K_{sp} = 1.6\times10^{-6}\right)$

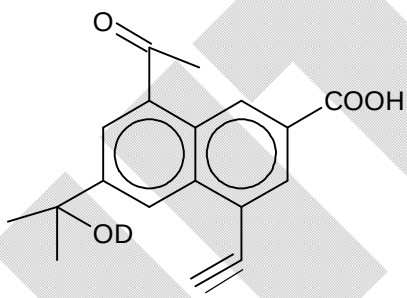
(C) $Pb(OH)_2$ $\left(K_{sp} = 2.8\times10^{-16}\right)$

(D) $Mg(OH)_2$ $\left(K_{sp} = 1.5\times10^{-10}\right)$

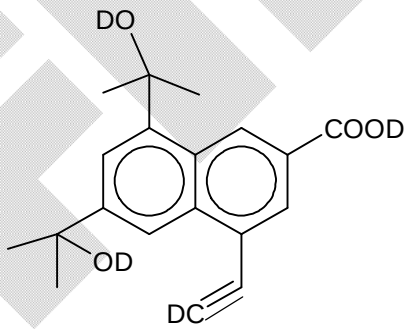
32. Final product of the reaction will be



(A)

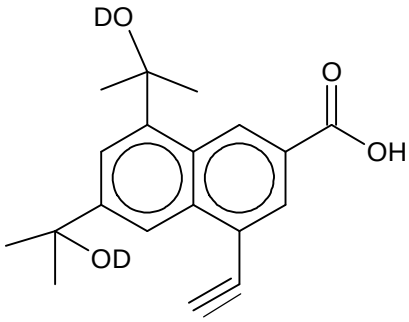


(B)

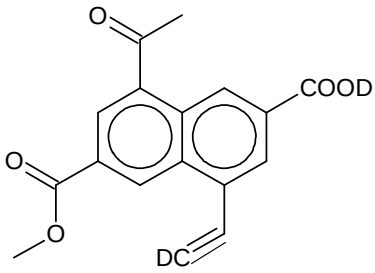


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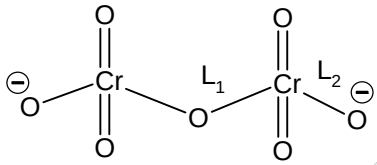
(C)



(D)



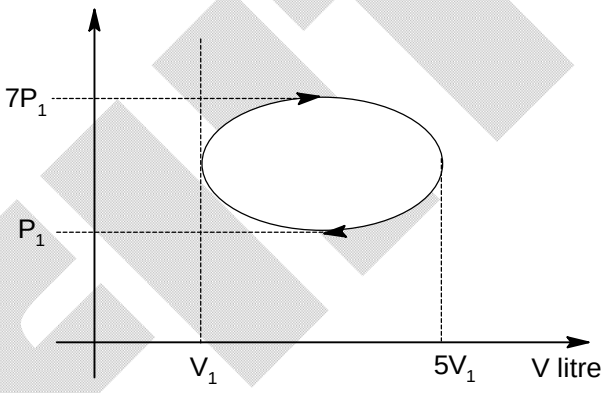
33. In $Cr_2O_7^{-2}$ structure L_1 and L_2 are the bond length of Cr – O then compare L_1 and L_2



(A) $L_1 = L_2$ (B) $L_1 > L_2$

(C) $L_1 < L_2$ (D) None of these

34. Find out the magnitude work done of given cyclic process as given in diagram



(A) $6\pi P_1V_1$ (B) $35\pi P_1V_1$

(C) $24\pi P_1V_1$ (D) $12\pi P_1V_1$

35. $A \rightarrow D$, reaction is zero order. Half-life of this reaction is 100 min. The time taken for the concentration of reactant "A" to reduce to one eight of it's initial value is_____

(A) 175 min (B) 400 min

(C) 200 min (D) 225 min

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36. $A(g) + 2B(g) \rightleftharpoons 3D(g) + 4C(s)$

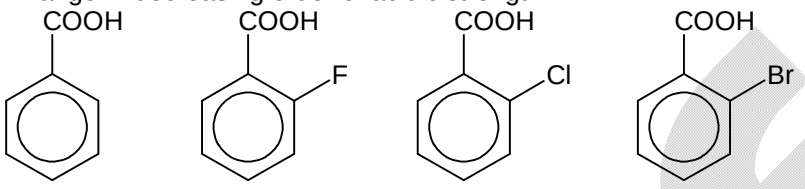
Find out unit of K_c for the reaction:

(A) M
(B) M^{-1}
(C) M^4
(D) None of these

37. A cylinder with movable piston filled with water which is in equilibrium with water vapour at 35°C. Identify by which condition water vapour will decrease

(A) Removing some water vapour
(B) Removing some water
(C) Moving piston upward at same distance
(D) Adding some NaCl in water

38. Arrange in decreasing order of acidic strength



(i) (ii) (iii) (iv)

(A) ii > iii > iv > i
(B) iv > iii > ii > i
(C) i > ii > iii > iv
(D) iii > ii > iv > i

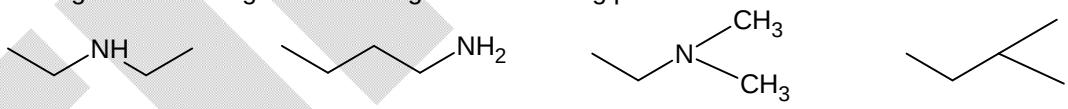
39. Identify the **incorrect** match in the following

	Column A (Vitamines)	Column B (Deficiency diseases)
(A)	A	Xerophthalmia
(B)	B_1	Cheilosis
(C)	B_6	Convulsions
(D)	D	Osteomalacia

40. Order of 1st ionization enthalpies of Cu, Co, Ni, Zn

(A) Zn > Co > Cu > Ni
(B) Ni > Zn > Cu > Co
(C) Zn > Cu > Co > Ni
(D) Zn > Ni > Co > Cu

41. Arrange the following in decreasing order of boiling points:



(i) (ii) (iii) (iv)

(A) i > ii > iii > iv
(B) ii > iii > i > iv
(C) ii > i > iii > iv
(D) iii > i > ii > iv

42. Arrange the following aqueous solution in decreasing order of freezing point. (Assume $\alpha = 1$ for all salts).

(i) 2 M NaCl
(ii) 5 M $CaCl_2$
(iii) 3 M Na_3PO_4
(iv) 2 M $Ca_3(PO_4)_2$

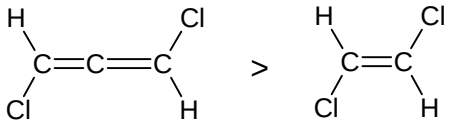
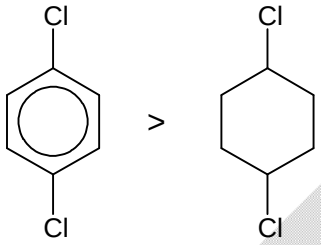
(A) i > ii > iii > iv
(B) iv > iii > ii > i
(C) ii > iii > i = iv
(D) i > iv > iii > ii

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43. Which of the following will show lilac colour on flame test.
(A) Potassium (B) Barium
(C) Copper (D) Zinc

44. Both optical and geometrical isomerism can be shown by
(A) $\left[Cu(en)(NH_3)_2\right]^{+2}$ (B) $\left[Co(en)_3\right]^{+3}$
(C) $\left[Co(en)_2(NH_3)_2\right]^{+3}$ (D) $\left[Co(en)(NH_3)Cl\right]^{+2}$

45. AgBr is insoluble in which of the following
(A) HNO_3 (B) NH_3
(C) $Na_2S_2O_3$ (D) NaCN

46. Identify correct order for dipole moment
(A) $CH_3F > CH_3Cl$
(B) 
(C) $NF_3 > NH_3$
(D) 

47. Identify oligosaccharides in the following
(A) Ribose (B) Gums
(C) Maltose (D) Glycogens

48. Chromate ion on reaction with acidified H_2O_2 give blue solution which turns green on standing blue colour appear due to formation of
(A) CrO_4^{-2} (B) CrO_5
(C) $Cr_2O_7^{-2}$ (D) Cr_2O_3

49. $E^o_{reduction}$ for metal M is given as in volts
$E^o_{M^{+3}/M^{+2}} = x$ $E^o_{M^{+3}/M} = y$ $E^o_{M^{+2}/M^{+1}} = z$
Value of $E^o_{M^+/M}$ will be
(A) $3y - x - z$ (B) $\frac{x - z}{3y}$
(C) $y - x - z$ (D) $x + 3y + z$

50. Correct order for the wavelength of absorption in the visible region for the following:
(A) $\left[CoCl_6\right]^{-3} > \left[Co(NH_3)_6\right]^{+3} > \left[Co(H_2O)_6\right]^{+3}$
(B) $\left[Co(NH_3)_6\right]^{+3} > \left[Co(H_2O)_6\right]^{+3} > \left[CoCl_6\right]^{-3}$
(C) $\left[CoCl_6\right]^{-3} > \left[Co(H_2O)_6\right]^{+3} > \left[Co(NH_3)_6\right]^{+3}$
(D) $\left[Co(H_2O)_6\right]^{+3} > \left[Co(NH_3)_6\right]^{+3} > \left[CoCl_6\right]^{-3}$

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SECTION – B

(Numerical Answer Type)

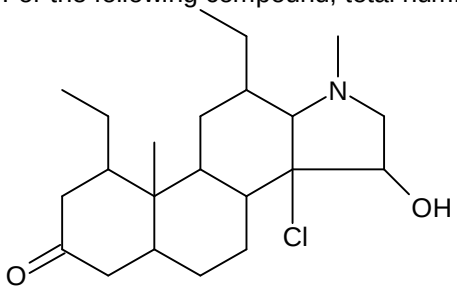
This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. How many elements follow $4d^{1\text{ to }10}\ 5s^{1}$ electronic configuration in fifth period of transition elements

52. In mischmetal alloy, what is approximate percentage of Fe present?

53. The ratio of mass percentage of "C and H" and "C" and "O" of a unsaturated acyclic organic compound 'Y' are 6 : 1 and 9 : 4 respectively. Then the moles of oxygen gas required for complete combustion of three molar of organic compound "Y" is_____

54. For the following compound, total number asymmetric atoms are



55. If wavelength of second line of Balmer series of hydrogen like species is 175 nm, then the wavelength of first line of Paschen series.

56. 20 gram of NaOH was dissolved in deionized water to prepare a 500 ml stock solution. What volume (in ml) of this solution would be required to prepare 1000 ml of 0.2 M solution?

57. Find out maximum number of electron which have $n = 5,\ \ell = 4$ and $s = \pm\frac{1}{2}$ for an element.

58. If equilibrium constant for the formation of NO_2 from NO and O_2 at 300 K is 10^{-x}.

$$NO(g) + \frac{1}{2}O_2(g) \rightleftharpoons NO_2(g)$$

At 300 K

$\Delta G^{o}_{(NO_2)} = 5.1454$ Kcal / mole , $\Delta G^{o}_{(O_2)} = 0$

$\Delta G^{o}_{(NO)} = 1.0$ Kcal / mole

Find the value of x?

59. Consider from Gr. 14 to Gr. 17, find out $x \times y$, where

x = total number of metalloids

y = total number of non-metals in periodic table.

60. In period 4, find out the atomic number of d-block element which works as oxidizer in organic chemistry reactions.

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Mathematics

PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. Let $f:\left[-\frac{3}{4},3\right]\to R$ and $g:\left[-\frac{3}{4},3\right]\to R$ be two functions defined by f(x) = $[x^2 - 5]$; ([t] represents greatest integer less than or equal to t and g(x) = $|2x - 3|$ f(x) + $|5x - 12|$ f(x). If f is discontinuous exactly at m points in $\left[-\frac{3}{4},3\right]$ and g is non differentiable exactly at n points in $\left(-\frac{3}{4},3\right)$ then m + n is equal to

(A) 16 (B) 17
(C) 18 (D) 20

62. The number of distinct real roots of the equation
$x^5(x^3 - x^2 - 4x + 4) + x(2x^3 - 5x^2 - 5x + 20) - 12 = 0$
is/are

(A) 0 (B) 3
(C) 6 (D) 8

63. If m and n be the absolute maximum and minimum values of the function
f(x) = $|x^2 - 7x + 10| - 5x + 27$; $x \in [-2, 14]$
then m + n is

(A) 65 (B) 66
(C) 68 (D) 70

64. If $\lim\limits_{x\to 0}\frac{Axe^x + B\log_e(1-x) + Cx^2e^{-x}}{x\tan^2(x/\sqrt{2})} = 2024$, then the value of A + B + $\frac{4C}{3}$ is

(A) 506 (B) 1012
(C) 2024 (D) 1518

65. Let $J(x) = \int e^{\tan^2 x}\left(\frac{2\tan^3 x}{1+\tan^2 x}\right)dx$, J(0) = 1, then $J\left(\frac{\pi}{4}\right)$ is

(A) e (B) $\frac{e}{2}$
(C) $\frac{e}{\sqrt{2}}$ (D) –e

66. Consider a chord PQ of the parabola $y^2 = 4ax$, a > 0. The tangents to the parabola at P and Q make an angle $\frac{\pi}{4}$ with the line y = 2x + 7 and cuts an intercept of length 12.5 units on y-axis. Then the length of latus-rectum of parabola is

(A) 10 (B) 15
(C) 25 (D) 50

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67. Let α and β be complex numbers with imaginary part 1 such that for all complex numbers $z = x + iy$ satisfying $\text{Arg}\left(\frac{z-\alpha}{z-\beta}\right) = \frac{\pi}{4}$, the ordered pair (x, y) lies on the circle $x^2 + y^2 - 8x - 6y + 17 = 0$, then $|\alpha\beta|$ is

(A) $\sqrt{175}$ (B) $\sqrt{185}$

(C) $\sqrt{195}$ (D) 15

68. Let $y = mx + c$ be a line passing through the point (3, 4) and making an intercept of length $\sqrt{5}$ units between the lines $y = 3x + 10$ and $y = 3x + 5$. The sum of all possible values of m is

(A) $\frac{1}{2}$ (B) -2

(C) $\frac{5}{2}$ (D) $-\frac{3}{2}$

69. The remainder obtained when $37^{2024} + 43^{2024}$ is divided by 16 is

(A) 0 (B) 2

(C) 3 (D) 7

70. Let A be the set consisting of the first 2024 terms of arithmetic progression 1, 5, 9, 13, ….. and B be the set consisting of the first 2024 terms of the arithmetic progression 3, 8, 13, 18, …… Then the number of elements which belongs to exactly one of the set A or B is/are

(A) 4048 (B) 3643

(C) 3238 (D) 3600

71. Consider an ellipse E : $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ having eccentricity $\frac{\sqrt{5}}{3}$; a > b. Let the tangent at any point P cuts the coordinates axes at points Q and R respectively. If minimum length of the intercept QR is 5, then length of latus-rectum is

(A) 9 (B) 8

(C) $\frac{8}{3}$ (D) $\frac{3}{2}$

72. If the line $2x - y + 13 = 0$ is a tangent to the curve $\frac{x^2}{a^2} - \frac{y^2}{27} = 1$, then the sides of a right angled triangle can be

(A) 3, 4, a (B) 4, 10, 2a

(C) 2a + 1, 2a + 3, 8 (D) a, 8, 25

73. The area of the region $S = \{(x, y) : (x - 2024)[x] \le y \le 4\sqrt{x};\ 0 \le x \le 2\}$; (where [k] represents greatest integer function) is

(A) $\frac{16\sqrt{2}}{3}$ (B) $\frac{16\sqrt{2}}{3} + 2022$

(C) $\frac{16\sqrt{2}}{3} + \frac{4045}{2}$ (D) $\frac{8\sqrt{2}}{3} + \frac{4045}{2}$

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74. If $\sum_{k=1}^{9}\left[\sec\left(\theta+(2k-1)\frac{\pi}{8}\right)\sec\left(\theta+(2k+1)\frac{\pi}{8}\right)\right]=\left(2+\frac{2}{\sqrt{3}}\right)\sqrt{2}$ has n distinct solutions in $\left(0,\frac{\pi}{2}\right)$, then sum of all the solutions is

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$

(C) $\frac{\pi}{8}$ (D) $\frac{\pi}{2}$

75. For x, y $\in$ R, let $\tan^{-1}x\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ and $\cot^{-1}y\in(0,\pi)$, then the sum of all the solutions of the equation $\tan^{-1}\left(\frac{10x}{25-x^2}\right)+\cot^{-1}\left(\frac{25-x^2}{10x}\right)=\frac{\pi}{3}$; 0 < |x| < 5 is equal to

(A) $10-15\sqrt{3}-\frac{5}{\sqrt{3}}$ (B) $7\sqrt{10}-15\sqrt{3}+\frac{5}{\sqrt{3}}$

(C) $10-5\sqrt{3}-\frac{5}{\sqrt{3}}$ (D) $10\sqrt{7}+15\sqrt{3}-\frac{5}{\sqrt{3}}$

76. If in triangle ABC, BC = 5 units, $\angle B = \sin^{-1}\frac{(\sqrt{3}+1)}{2\sqrt{2}}$ and radius of circum circle of ΔABC is $\frac{5}{\sqrt{2}}$ units, then area of ΔABC is

(A) $\frac{25\sqrt{3}(\sqrt{3}+1)}{4}$ (B) $\frac{25\sqrt{3}(\sqrt{3}+1)}{2}$

(C) $\frac{25\sqrt{3}(\sqrt{3}+1)}{8}$ (D) $\frac{25(\sqrt{3}+1)}{8}$

77. If $32\sin^5\alpha + 243\cos^5\beta + 3 = 30\sin\alpha\cos\beta$; $\alpha, \beta \in \left[0,\frac{\pi}{2}\right]$, then the value of $\cos2\alpha + \cos2\beta$ is

(A) $\frac{5}{18}$ (B) $-\frac{5}{18}$

(C) $-\frac{7}{9}$ (D) $\frac{7}{9}$

78. Let R be a relation on set of real numbers given by R = {(a, b) : pa – pb + q is irrational ; p is rational q is irrational), then R is
(A) reflexive and symmetric but not transitive
(B) reflexive and transitive but not symmetric
(C) reflexive but neither symmetric nor transitive
(D) an equivalence relation

79. If $\int\frac{\cos x}{\sqrt{x^2-\sin^2 x}}dx-\int\frac{\sin x}{x\sqrt{x^2-\sin^2 x}}dx=\text{cosec}^{-1}(g(x))+c$, then $\lim_{x\to0}\{g(x)\}$ is ({t} represents fractional part of t)

(A) does not exist (B) 1

(C) 0 (D) –1

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80. Let y = y(x) be the solution of the differential equation $\frac{dy}{dx} = \left((y+2024)e^{x^3/3} - x^2\right)(y+2024)$; $-3 < x < 3$ with $y(0) = -\frac{6071}{3}$, then y(2) is

(A) $\frac{1}{e^{8/3}} - 2024$ (B) $2024 - \frac{1}{e^{8/3}}$

(C) $e^{-8/3}$ (D) $1 - \frac{1}{e^{8/3}}$

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. Let A = {1, 2, 3, 4, …….. 2024}, B = {4, 8, 12, 16, ……. , 2400}. If the number of possible sets C that can be formed where $C \subseteq A$ and $C \cap B \neq \phi$ is $(2^p - 1)2^m$, then sum of even divisors of (m – p) is/are

82. A rectangle R with end points of one of its sides as (3, 5) and (7, 9) is inscribed in a circle. If the equation of one of the diameters of circle is 3x – y – 12 = 0 then area of the rectangle is

83. The points A, B, C represented by the complex numbers z_1, z_2 and z_3 respectively lie on circumference of a circle drawn on OA as diameter ; O being origin. If $\angle BOC = 2\angle AOB$ and $z_2^3 = z_1^2 z_3\left(\frac{5+3\sqrt{3}}{8}\right)$, then $\angle BOC = \frac{\pi}{k}$ where k is

84. If α and β are the roots of the equation $x^2 - 13x - 1 = 0$, then the value of $\frac{\alpha^{2024} + \beta^{2024} + \alpha^{2020} + \beta^{2020}}{\alpha^{2022} + \beta^{2022}}$ is

85. The number of six digit natural numbers such that product of its digits is 24 is

86. Let A be a n × n matrix such that |A| = 3. If the determinant of the matrix $Adj(5.Adj(3A^{-1}))$ is $3^{216}.5^{42}$, then $\pi \int_{n-14}^{n} x \sin \pi x \, dx$ is

87. Let L_1 be a line in xy-plane with x and y intercepts 8 and 6 respectively and L_2 be a line in yz plane with y and z intercepts –8 and –5 respectively. If d is the shortest distance between L_1 and L_2 then $\frac{1649}{280}$ d^2 is

88. Let $\vec{a} = \frac{3\hat{i} + 4\hat{j} + 5\hat{k}}{5\sqrt{2}}$, $\vec{b} = \frac{5\hat{i} - 5\hat{j} + \hat{k}}{\sqrt{51}}$ be two given vectors, then the value of $(2023\vec{a} + 2024\vec{b}) \cdot \{(2\vec{b} - 3\vec{a}) \times (\vec{a} \times \vec{b})\}$ is

89. Let S = {1, 2, 3, 4, 5, 6, 7} if the probability that a randomly chosen onto function ' f ' defined from S to S satisfies the condition that 2f(2) – f(1) – f(3) = 0 is p, then the value of 2100p is

90. Let the mean of observations 1, 3, 4, 7, x, y be 6 and their variance is 14. If the mean deviation about the mean is k then the value of 60k is